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Functional Analysis in Virtual Environments

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Abstract

Functional analysis (FA) is an assessment procedure involving the systematic manipulation of an individual's environment to determine why a target behavior is occurring. An analog FA provides practitioners the opportunity to manipulate variables in a controlled environment and formulate a hypothesis for the function of a behavior. In previous studies, researchers have attempted to train practitioners who have no experience with FA to implement the conditions with procedural integrity. In these studies, researchers used a traditional professional development model (e.g., didactic training, role-play). To date, simulation and virtual environments have not been used to enhance FA training procedures in the preparation of teachers and behavior analysts. In the current investigation, we sought to train educators to implement two test conditions of an analog FA (i.e., attention, escape). The intervention package included a didactic training and individual and group-rehearsed FA trainings in a virtual environment (i.e., TeachLivE™). Teachers demonstrated an increase in their fidelity of implementation within three sessions and generalized performance to a classroom setting. The implications for rural teachers are considerable as the results suggest we can train teachers on complex assessment techniques at a distance.

Keywords

functional analysis, functional behavior assessment, applied behavior analysis, behavior, simulation, virtual environments, professional development

A practitioner's ability to understand and manage student behavior in the classroom is critical to his or her success in becoming a highly effective practitioner (Garland, Garland, & Vasquez, 2013). Reschly and Holdheide (2008) found that teachers who are skilled in scientifically based instruction, classroom organization, and behavior management were able to establish classroom environments conducive to learning and improved academic performance for all students. Chief among the challenges facing special educators are problem behaviors that students display in classroom environments. The prevalence of challenging behaviors is significantly elevated in children with disabilities (Emerson et al., 2001; Murphy, Healy, & Leader, 2009). Specifically, about half of children with autism spectrum disorder (ASD) will display problem behavior (Hagopian, 2007).

Teachers who are able to analyze and find patterns in these challenging behaviors will likely be more successful in modifying them. A functional analysis (FA) is a procedure for evaluating the function of an individual's behavior and is considered the most accurate methodology for completing a functional behavior assessment (FBA; Beavers, Iwata, & Lerman, 2013; Iwata, Dorsey, Slifer, Bauman, & Richman, 1994). For example, Iwata, Pace, and colleagues (1994) found that, of 152 FAs completed over 11 years, 95.4% yielded usable results for treatment planning.

An analog FA requires the systematic manipulation of controlled environmental variables and is completed in a controlled setting, such as an assessment or treatment room, to limit the interference of outside variables and increase experimental control (Carr & Durand, 1985; Iwata, Dorsey, et al., 1994). These manipulations are completed to precisely identify the maintaining consequence or consequences, associated with a specific target behavior (e.g., aggression, stereotypy). By executing a series of analog conditions, practitioners are able to identify relevant reinforcement contingencies and formulate a functional hypothesis of maintaining variables (i.e., situations or factors associated with the problem behavior). These contingencies are considered representative of those occurring within the natural environment.

The Individuals With Disabilities Education Improvement Act (IDEA; 2004) stated that schools teams must address the behavioral needs of students when they impede learning. As a

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result, most teacher preparation programs include coursework on behavior management but may not be able to fully train teachers to conduct FBAs or FA procedures, thus leaving them without a solid foundation on which to construct effective behavior intervention plans. If practitioners do not have a clear sense of the maintaining variables prior to designing and implementing a behavioral intervention, then it is likely that the intervention will not be effective in modifying the target behavior. Several possible reasons why teacher preparation programs avoid teaching FA procedures include lack of trained practitioners, misconception of FA procedural difficulty, questions about feasibility in the school environment, and lack of resources (Gable, Tonelson, Sheth, Wilson, & Park, 2012).

Several studies have been published suggesting lay individuals with limited training in behavioral assessment can be instructed to implement FA procedures. Iwata and colleagues (2000) showed that upper level undergraduate students could be trained to implement the attention, control, and demand conditions of a FA. The researchers used a multiple baseline across subjects design. Baseline sessions consisted of reading the Iwata, Dorsey, et al. (1994) method, followed by participants implementing the two or three analog conditions based on their understanding of the procedures from the reading alone. For the intervention phase, Iwata and colleagues (2000) provided participants with both verbal and written descriptions of the conditions, followed by a video of an individual modeling correct implementation of each of the conditions. The researchers then asked participants to implement the conditions again and provided trainees with feedback. After intervention, all participants implemented the FA conditions with a high level of integrity.

Moore and colleagues (2002) demonstrated that three teachers could be trained to implement the demand and attention conditions of a FA. The researchers also used a multiple baseline across subjects design. For their intervention phase, Moore et al. delivered verbal and written descriptions, provided both a live demonstration for each of the FA conditions and gave participants the opportunity to rehearse the conditions prior to reassessment of implementation. Again, all participants demonstrated high procedural fidelity following intervention.

Similarly, Wallace, Doney, Mintz-Resudek, and Tarbox (2004) found that two teachers and a school psychologist could be trained to implement attention, control, and demand conditions. Again, the researchers used a multiple baseline design with a similar baseline phase. For the intervention phase, Wallace et al. (2004) used a multicomponent training that included descriptions of the conditions, a video model of the conditions, and an opportunity to rehearse the conditions. If the participants failed to implement the procedure with fidelity during the intervention phase, verbal feedback was provided. All participants implemented the FA conditions with a high level of integrity following training. In addition, in two of the studies, researchers demonstrated that the increased procedural integrity could be generalized to FAs that were conducted with real clients in a more naturalized setting (Moore et al., 2002; Wallace et al., 2004).

More recently, Bloom, Iwata, Fritz, Roscoe, and Carreau (2011) explored one alternative that is known as a trial-based FA. The trial-based FA can be conducted within the context of the student's typical daily activities in a more naturalistic environment. This variation of the FA methodology is slightly less accurate than a traditional FA in determining the function of a target behavior, but the trial-based FA is still more accurate that other descriptive measures that typically are used in schools. Therefore, the trial-based FA is a good alternative when resources (e.g., time and space) are more limited.

Kunnavatana, Bloom, Samaha, and Dayton (2013) extended the work of Bloom and colleagues (2011) and trained four teachers to implement trial-based FAs using didactic teaching and practice with feedback. The didactic portion of the training included an introduction to standard FA and review of trial-based FA procedures, as well as video models of each FA condition (i.e., ignore, attention, escape, tangible). The practice with feedback portion of the training included small group practice with immediate feedback for any errors and an opportunity for participants to ask questions followed by "test sessions" where the participants implemented each FA condition with a graduate student acting as the subject. Each participant implemented the four FA conditions (i.e., ignore, attention, escape, and tangible) and received feedback at the end of each session. Participants performed the conditions until they implemented them with 100% procedural fidelity. These researchers also investigated whether the participants' procedural fidelity generalized into the classroom environment 1 to 3 months after the initial training. For all participants, performance remained above baseline levels in this generalization phase.

Finally, Alnemary, Wallace, Symon, and Barry (2015) used a protocol to train four international teachers with no FA experience to implement four FA conditions (i.e., play, ignore, attention, and demand) using video conferencing. The training included a 3-hr group training with readings, video modeling, and role-play followed by individualized feedback. The researchers found that all four participants were able to reach mastery criteria for two conditions, and only one participant was able to meet mastery criteria for all four FA conditions.

To date, simulation has not been explored as an option for training practitioners in FA procedures but may offer a more realistic rehearsal experience as compared with role-play with a colleague or graduate student. Simulation training has been a developing field that offers an interactive environment for participants, while providing researchers a range of performance data (e.g., knowledge, skills, heart rate; Moskaliuk, Bertram, & Cress, 2013; Noteborn, Bohle Carbonell, Dailey-Hebert, &

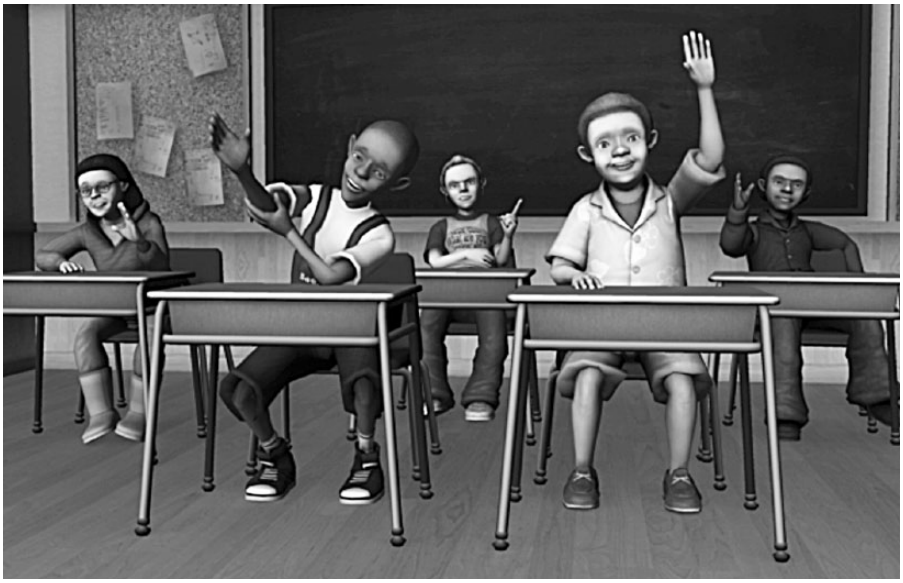


Figure 1. Image of the TeachLivE™ simulated environment.
Source. Reprinted with permission from TeachLivE Group (http://teachlive.org).

Gijselaers, 2012; Salas, Wildman, & Piccolo, 2009). Although training in the classroom environment is of great importance, often it is not cost effective, and in some situations (e.g., industrial training), could be dangerous (Bertram, Moskaliuk, & Cress, 2015; Peeters, van den Bosch, Meyer, & Neerincx, 2014; Stedrak, Ortagus, & Wood, 2012). When teaching untrained teachers or practitioners to run FA procedures with real students or clients, there are some ethical issues, such as lack of knowledge about when to discontinue the procedure and possible safety issues when conducting the procedure with an individual with severe aggression or self-injury. Simulation training in a virtual environment provides researchers and participants a safe environment to practice skills and tasks (Hahn, 2010; Salas et al., 2009) and provides a proven base for knowledge acquisition and transfer (Moskaliuk et al., 2013).

TeachLivE™ is a mixed-reality teaching environment supporting teacher practice in pedagogy and content (see Figure 1). The innovative technology of this lab affords participants the opportunity to build content area and pedagogical skills with an understanding of how skills will transfer to a real classroom situation. Dieker, Hughes, Hynes, and Straub (2013) demonstrated that four 10-min sessions in the TeachLivE™ simulator resulted in the transfer of at least one teacher behavior back to the "real" classroom. TeachLivE™ is being used by universities around the country to develop specific skills in educators, including use of open-ended questions, discrete trial training, reinforcer sampling, and classroom or behavior management strategies (Dieker, Rodriguez, Lingnugaris-Kraft, Hynes, & Hughes, 2014). This simulated environment may offer a controlled setting to instruct pre-service and in-service teachers to not only implement the procedures but also collect and analyze the data generated by the assessment. Therefore, the purpose of this study is to replicate procedures from Kunnavatana and colleagues (2013) and extend the literature by using the innovative virtual learning environment platform of TeachLivE™ as a method to practice FA procedures with feedback for pre-service teachers. The goal of this study is to determine how these skills generalize into the classroom environment.

Previous studies have utilized a training protocol for FA that involves a didactic component, a role-playing or practice component, and a posttraining assessment (Lambert, Bloom, Clay, Kunnavatana, & Collins, 2014; Lambert, Lloyd, Staubitz, Weaver, & Jennings, 2014; McKenney, Waldron, & Conroy, 2013). Teachers and other practitioners have generally shown the ability to implement a traditional FA or trial-based FA procedure with fidelity given this training protocol. The current investigation seeks to expand upon this body of research and investigate whether using simulation in place of the role-play component of the training will improve the efficiency and efficacy of the training protocol for in-service teachers in a master's program in urban, suburban, and rural settings. The implications for rural teachers are notable given the limited access to training centers and clients for professionals who are seeking out this specialized training.

Method

Setting

The researchers conducted all sessions at the Synthetic Reality Lab (SREAL) at the Institute for Simulation and Training (IST) on the University of Central Florida campus.

IST is equipped with the virtual reality and simulation technology TeachLivE™. The University of Central Florida has used this technology extensively for teacher preparation, medical training, and business training. The IST houses a TeachLivE™ lab on-site along with several rooms to provide anonymity and separation of participants so that contamination (e.g., participants viewing the sessions of other participants) does not occur.

The TeachLivE™ lab has a large television screen on one wall, which features the virtual classroom and student avatar. There is space in front of the screen for the participant to move around in front of the avatar and chairs around the perimeter of the room for observers. The virtual student used for this study presented with an ASD, and participants were able to interact with the virtual student and have him or her respond in real time.

Materials

The researchers developed a task analysis for each test condition to operationalize the teacher behaviors and standardize the data collection for each condition. An example of the task analysis prompt for the attention (control) condition stated, "Teacher provided continuous attention (no more than 10 seconds between interactions) to the student until the student engaged in target behavior or until one minute elapsed." An example of the attention (test) condition stated, "Teacher turned away from the student and stopped providing attention (and did not issue any demands)." An example of the task analysis prompt for the escape condition (control) stated, "Teacher DID NOT deliver any demands." An example of the task analysis prompt for the escape condition (test) stated, "Teacher provided instruction and prompts (including model and physical, if relevant) without delays over 10 seconds between demands, prompts, or ongoing work."

Participants

Three public school teachers, two male and one female, enrolled in a master's-level program at the University of Central Florida participated in the study. All participants had at least 1 year of classroom teaching experience. Sam had 1 year of classroom experience, prior exposure to the TeachLivE™ simulator, and no experience with FA. Sam self-reported high confidence with behavior management concepts and strategies based on informal interview. Samantha had 2 years of classroom experience, previous exposure to the TeachLivE™ simulator, and no experience with FA. This participant reported a high level of confidence with behavior management. Dyan had 1 year of classroom experience, no previous exposure to the TeachLivE™ simulator, and no FA experience. Dyan reported limited confidence in her classroom behavior management skills.

The researchers recruited the participants via email to all students in a master's degree–level course at the University of Central Florida in the College of Education. They provided interns with general information about the study, including that it involved implementing FA procedures in the TeachLivE™ simulator. Participants who met the following criteria were eligible for the study: (a) no previous experience with FA and (b) currently working in a classroom setting with students who display some maladaptive behaviors.

Experimental Design and Procedure

The researchers used a multiple baseline across participants design to assess the effects of training on correct implementation of the attention and escape conditions. They collected data on the teachers' fidelity of implementation of FA test conditions across both baseline and intervention phases (Erbas, Tekin-Iftar, & Yucesoy, 2006; Iwata, Dorsey, et al., 1994; Kunnavatana et al., 2013). Each teacher completed both conditions, and the researchers randomized the sequence of the conditions. Similar to Bloom et al. (2011), the researchers segmented FA conditions into a control and test trial that lasted up to 5 min. The purpose of both trials was to gauge whether the problem behavior existed in both or either trial to determine the function of the behavior.

Analog Conditions

Two test conditions (i.e., attention, escape) occurred during baseline and intervention phases. These conditions determined the function of the behavior, replicating Iwata, Dorsey, et al. (1994), Erbas et al. (2006), and Kunnavatana et al. (2013). Teachers demonstrated their ability to execute each condition with skills required for FA. Each teacher completed the same number of conditions, and the researchers randomized the sequence of each condition.

Attention condition. The attention condition had two components: control and test segments. The control segment lasted for 1 min during which the students received ongoing attention (e.g., "Wow, I really like your outfit!"). A 2-min test segment followed this condition when attention was not provided (i.e., turning away from the student in question and delivering attention to another student).

Escape condition. The escape condition also included control and test segments. First, during the 1-min control segment, the student received no demands (e.g., "You can put your head down on your desk"). Next, during the 2-min test segment, the teacher placed ongoing demands on the student (e.g., "Return to working at your desk" or "Pick up your pencil").

Baseline

The researchers provided the teachers with graphs of hypothetical FA data and multiple-choice questions related to the analysis of these data. This baseline assessment consisted of 10 questions and focused on the participants' understanding or interpretation of hypothetical FA case studies. The researchers then asked the participants to conduct the FA conditions with a student avatar in the simulator. Sessions were 3 min in length, and an interactor assumed the role of an avatar that engaged in verbal tantrums (target behavior) and other behaviors. Interactors followed scripts that specified the times at which maladaptive behaviors should occur. The researchers gave the participants the "Methods" section of the Iwata, Dorsey, et al. (1994) study and told them to review it prior to conducting these sessions. They also asked the teachers to collect data as they implemented sessions. The researchers gathered procedural integrity data during the two different trial types. During baseline condition, data collectors and the primary researcher provided no feedback to the participant.

Training

The training portion of this study was comprised of two separate segments: (a) a lecture on the fundamentals of applied behavior analysis (ABA) and FA, and (b) a small group role-play with corrective feedback. A Board Certified Behavior Analyst (BCBA) delivered the lecture component of the training. The participants did not have any formal experience in ABA, so this provided exposure to behavioral principles (e.g., reinforcement, punishment), FA procedures, and case study practice. This training included a description on the link between antecedents, behaviors, and consequences to help illustrate the importance of identifying the function of a behavior when developing a behavior intervention plan. Case studies provided a chance for participants to learn how to begin assessing the function of a target behavior. A follow-up session extended the didactic portion of this presentation to include a data collection practice opportunity for the purposes of the FA. The researchers instructed the participants on how to fill out the data collection sheets for each condition in the FA trials.

The second portion of the training consisted of small group role-play. A doctoral candidate with her BCBA modeled the process for performing in each trial type. The primary researcher acted as a student with maladaptive behaviors, and the participants practiced how to correctly respond to the student and code data in each trial type. The facilitator provided immediate feedback on the performance of participants. The researchers asked the participants to complete a content knowledge exam and pass at 90% or higher prior to moving into the last phase of the study.

Measurement and Reliability

The researchers collected all data in the simulator. In situ (i.e., classroom) measurement was not available given that the participants were on summer break working as externs at the University of Central Florida. The dependent variable was the procedural integrity of the respective FA trials (i.e., attention and escape). The researchers defined procedural integrity as the percentage of steps performed correctly in the task analysis of each condition. They adapted ask analyses from Kunnavatana et al. (2013) for the two trial types and scored accordingly: (a) occurred, (b) did not occur, or (c) not applicable. They calculated procedural integrity by dividing the number of correct items in the task analysis by the number of total items in the task analysis and multiplying by 100.

Second and third observers were present for all trials of baseline and intervention to collect data independent of the primary observer. Interobserver agreement (IOA) yielded a reliability figure by dividing the number of agreements by the number of agreements plus disagreements and then multiplying by 100. The researchers calculated IOA at 94% across all three observers with a range from 90% to 98%. Prior to coding any data, the researchers required all observers to watch a video of each FA condition being implemented and collect data on the frequency of target behaviors. All observers achieved 100% accuracy prior to coding any study data. The purpose of this IOA training was to ensure the accuracy of the data and to prevent observer drift. The researchers implemented ongoing booster sessions during the course of the training to ensure the most reliable results. They did not record the sessions in the simulator due to confidentiality concerns.

The researchers also collected data on teachers' ability to analyze and graph data from hypothetical FAs. As per Kunnavatana et al. (2013), they asked the participants to create a bar graph based on the FA data with a paper-and-pencil chart. They also gave the participants five hypothetical bar graphs and asked them to answer multiple-choice questions based on the FA data. They offered a postintervention opportunity to complete these graphs at the completion of the simulator portion of the study.

Results

The purpose of this study was to train teachers in FA using a virtual simulator (i.e., TeachLivE™). The primary research question was as follows:

> **Research Question 1:** To what extent does the combination of simulation and FA training increase the procedural fidelity of FA procedures in TeachLivE™ with in-service teachers?

Figure 2 shows the teachers' procedural integrity (i.e., the percentage of steps performed correctly). Baseline data varied

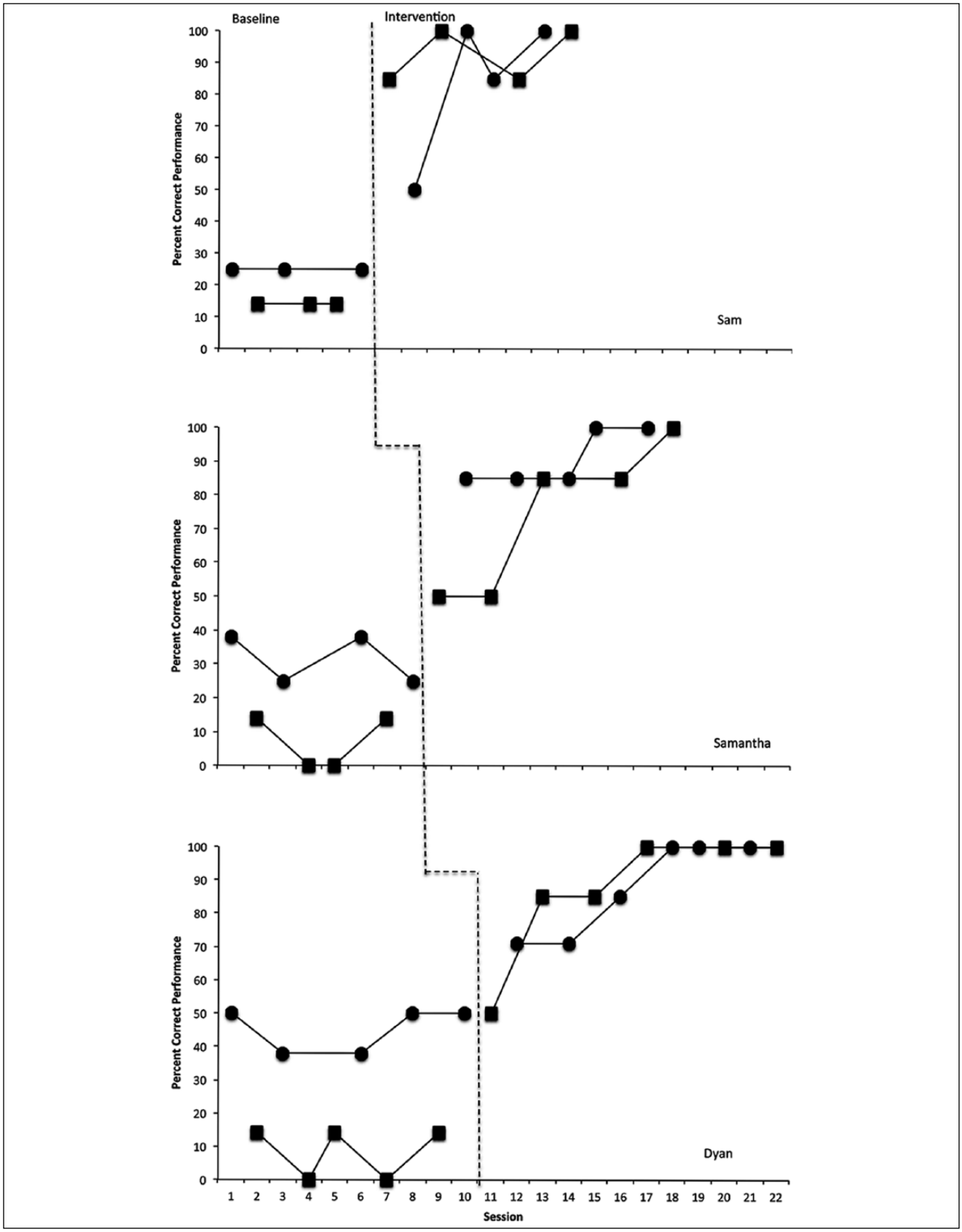


Figure 2. Multiple baseline outcomes for teacher fidelity of implementation of FA procedures in simulated environments.
Note. FA = functional analysis.

from 14% to 50% accuracy in attention and escape trials. During baseline condition, Sam conducted the attention trials with 25% and the escape trials with 14% accuracy. Samantha conducted the trials with 38% and 14%, respectively. Dyan performed at 50% and 14%, respectively.

Upon completion of the two training segments, the procedural integrity for all participants increased. After the first phase of training, Sam conducted the attention trials with 50% and then 100% procedural integrity. His escape trials also improved from 85% to 100%. Samantha increased to 85% in attention and 50% in escape. Dyan improved from 71% to 100% accuracy in the first postintervention phases for attention and 50% to 100% in the escape conditions. After the second training phase, all participants reached 100% accuracy in both attention and escape trials. Ongoing feedback was instrumental in achieving this end. The researchers offered the participants numerous opportunities to ask questions and have corrective feedback provided during the role-play components. This increased all procedural integrity figures to 100%.

Discussion

Among the challenges that face special educators today are the frequency and intensity of the challenging behaviors that their students display in the classroom environment. The prevalence of challenging behaviors is significantly elevated in children with ASD and intellectual disabilities (Emerson et al., 2001; Murphy et al., 2009). The researchers designed this study to evaluate whether technology and simulation can enhance the efficacy of FA training for teachers. Baer, Wolf, and Risley's (1968) seminal work posited that it is important to carefully consider the goals, procedures, and outcomes as they relate to practice in ABA. Given that the goals and procedures related to FA have been established by Kunnavatana et al. (2013) and previous seminal leaders (Iwata, Dorsey, et al., 1994), it remains an important question whether a simulated platform could also deliver positive outcomes that generalize to clinical/real-world settings. Simulations allow instructors to provide repeated exposure to varying levels of intense behavior in a safe environment with no harm done to real children. The results of this research suggest all three teachers conducted trial-based FAs with competent procedural integrity after one training session and nearly total competence after a second training session.

These findings present a new world of opportunities for the training of teachers, behavior analysts, psychologists, and caretakers in learning the complex world of FA and ABA procedures. Further studies could extend this work in simulators to explore the generalization to classrooms or clinical settings. In addition, a large efficacious study would address the scalability of this platform for enhancing practitioner skills.

This intervention also might be modified to train teachers in other behavior analytic skills or strategies. For example, a protocol that incorporates simulation could be used to teach educators to use prompt fading techniques, implement individual contingencies (e.g., token boards), implement group or class contingencies, and collect behavioral data. Although this study focused on teachers as participants, a similar training package could be used with school psychologists, behavior specialists, or other behavior support staff in school or clinical settings.

Finally, there is consistent demand for BCBAs and Board Certified Assistant Behavior Analysts (BCaBAs) in school and clinical settings. To receive certification, behavior analysts are required to complete extensive, supervised fieldwork or practicum experiences. By using avatars and virtual environments as a part of this training, practitioners could have opportunities to implement procedures, such a preference assessments, imitation training, verbal behavior training (e.g., echoics, mands, tacts, intraverbals), high-probability request sequences, and errorless learning. Virtual environments and simulation have the potential to enhance these supervision experiences and assist in the preparation of highly qualified BCBAs and BCaBAs.

There are several limitations that should be considered when interpreting the results of the current analysis. First, this investigation included the low number of participants which limits the generalizability of the results. Second, after the initial didactic training, the researchers delivered all future training sessions in the simulator one at a time to individual participants. This may not be the most economical approach for school districts that wish to train large numbers of teachers, behavior specialists, and other providers on FA procedures. Future research should be conducted to evaluate whether this approach to FA training is effective when groups of teachers are trained simultaneously in virtual environments. Third, the researchers did not measure procedural fidelity in vivo (i.e., live classroom) with students in the classroom. Little is known about the generalizability of the simulator leading to questions about scalability. Finally, the researchers did not collect follow-up data; thus, it is unclear as to whether the skills teachers acquired and demonstrated were maintained after a period of time.

Declaration of Conflicting Interests

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